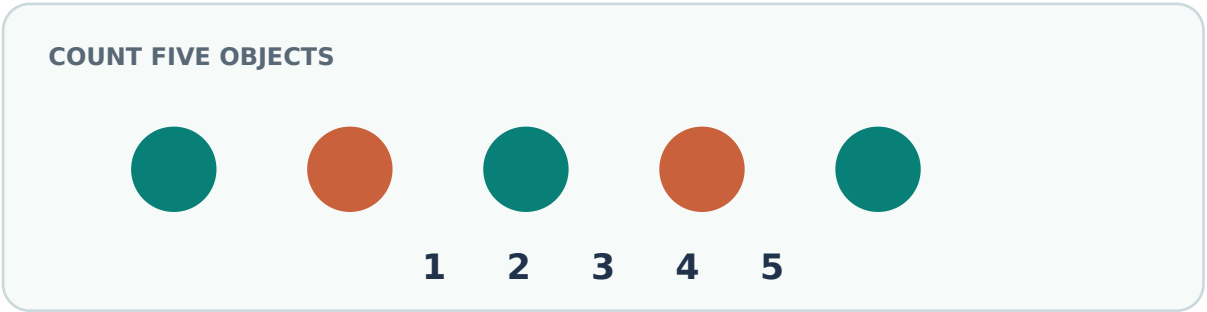
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MATHEMATICS BASIC 1

A Learner’s Guide

Discover numbers, patterns, shapes and useful maths through stories and activities.



PRINT EDITION • A4 • ORIGINAL CURRICULUM-ALIGNED CONTENT

Includes 24 lessons, hands-on investigations, more than 120 practice questions, revision, an answer guide and a number chart.

Welcome, young mathematician!

Mathematics helps us count, compare, share, measure and explain our world. This book is designed for Basic 1 learners studying the four Mathematics strands in the supplied NaCCA primary curriculum.

How to use this book

Read or listen to one lesson at a time. Discuss the examples with a teacher or trusted adult. Try the practical activities with safe objects. Answer the practice questions in a separate exercise book. Then check your work using the answer guide.

For teachers and parents

The lesson labels retain the curriculum learning-indicator codes. Learning is hands-on: let children explain their reasoning with real objects, drawings and speech. Adjust task difficulty and use the assessment questions as formative checks, not as high-pressure examinations.

Safety and printing

Small objects must never be put in the mouth. Water activities need adult supervision. Use paper money models instead of giving children real money to spend. Print on A4 paper at actual size; black-and-white printing will remain readable.

Curriculum source and independence

Based on the learning outcomes in Ghana Education Service / NaCCA, Mathematics Curriculum for Primary Schools (Basic 1-3), September 2019, Basic 1 section, source PDF pages 21-37. All learner explanations and questions in this book are original. This is an independently prepared educational guide, not an official NaCCA publication or endorsement.

Contents and curriculum map

01. Counting and naming numbers to 100 — B1.1.1.1.1

02. Finding numbers on a hundred chart — B1.1.1.1.2

03. Measuring by counting handspans and cups — B1.1.1.1.3

04. More, less and ordering numbers — B1.1.1.1.4

05. Greater than, less than and equal to — B1.1.1.1.5

06. Comparing numbers in everyday stories — B1.1.1.1.6

07. Understanding addition and subtraction — B1.1.2.1.1

08. Understanding equal groups and missing numbers — B1.1.2.2.1

09. Addition and subtraction are connected — B1.1.2.2.2

10. Making stories from number sentences — B1.1.2.2.3

11. Friendly number facts and doubles — B1.1.2.3.1

12. Smart ways to add and subtract within 20 — B1.1.2.4.1

13. One-step word problems — B1.1.2.4.2

14. Finding one-half — B1.1.3.1.1

15. Counting halves — B1.1.3.1.2

16. Ghanaian coins and their values — B1.1.4.1.1

17. Repeating patterns around us — B1.2.1.1.1

18. How we know a shape — B1.3.1.1.1

19. Solid shapes in the home and school — B1.3.1.1.2

20. Flat shapes and their properties — B1.3.1.1.3

21. Position and direction words — B1.3.2.1.1

22. Comparing length, mass and capacity — B1.3.3.1.1

23. Making simple picture graphs — B1.4.1.1.1

24. Counting, tallying and reading data — B1.4.1.1.2

Strand coverage

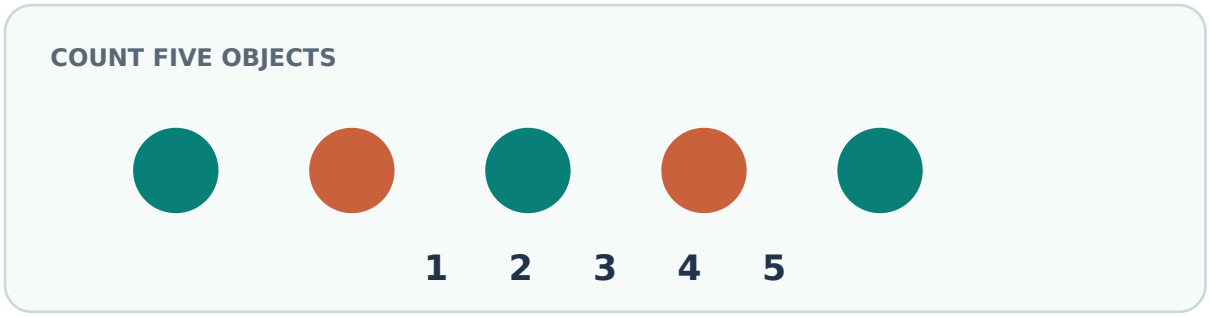
Number: Lessons 1-16 • Algebra: Lesson 17 • Geometry and Measurement: Lessons 18-22 • Data: Lessons 23-24.

LESSON 01 • NUMBER

Counting and naming numbers to 100

NaCCA indicator: B1.1.1.1.1

I can: Count forwards and backwards; count by twos and tens; match a group to its number; estimate and identify positions up to tenth.



Let’s learn

Numbers tell us how many things are in a group. We can count stones, books, mangoes, cups and pupils. Touch or point to each object once as you say its number name. The last number tells how many objects are there.

The number zero means there are none. To count forwards, say the next number: 16, 17, 18, 19, 20. To count backwards, say the number before: 20, 19, 18, 17, 16. A hundred chart helps us see numbers from 1 to 100.

Skip-counting is counting in jumps. When we count by twos, we may say 2, 4, 6, 8, 10. By tens, we say 10, 20, 30, 40, 50. Skip-counting helps when objects are in pairs or groups of ten.

An estimate is a sensible guess before we count exactly. You could guess that a handful contains about 15 bottle tops, then count to check. The positions first, second, third and so on to tenth are ordinal numbers; they tell us an order, not a total.

Worked examples

1. Count five chalk pieces: one, two, three, four, five. We write 5. Start at 24 and count by twos: 24, 26, 28, 30, 32.
2. A child is fourth in a line: three children stand ahead of the child. To estimate 24 seeds, group them in tens and notice two tens with a few extra.

Hands-on learning

1. Build groups of 10 beans on paper, then use the groups to count 10, 20, 30 and onwards. Do not put small objects in the mouth.
2. Stand in a line of up to ten children. Say who is first, second, third and last.

Practise in your exercise book

1. What is the number after 39?
2. Count backwards: 12, 11, __, 9.
3. Complete: 10, 20, 30, __, 50.
4. Count by twos: 14, 16, __, 20.
5. Which word names position: five or fifth?
6. You count eight pencils. Write the numeral.
7. What number means no objects?

Think further

Estimate how many steps it takes to cross a room, then count the steps and compare.

Try this at home

Count 20 safe objects in your home. Show how you arranged them so you did not count any object twice.

Quick self-check

1. Which comes after 59? A. 60 B. 58 C. 69
2. What comes next? 2, 4, 6, ... A. 7 B. 8 C. 10
3. Which word describes a place in line? A. third B. three C. thirty

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 02 • NUMBER

Finding numbers on a hundred chart

NaCCA indicator: B1.1.1.1.2

I can: Read numbers in a chart and tell which number is above, below, before, after or beside another.

A ROW OF TEN NUMBERS

21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**

Move right: +1 | Move down in a hundred chart: +10

Let’s learn

A hundred chart puts numbers in rows. In an ordinary chart with ten numbers in each row, moving one place right usually adds 1. Moving one place left usually subtracts 1.

Moving down one row adds 10: 24 becomes 34. Moving up one row subtracts 10: 46 becomes 36. These movements help us look for number patterns.

The words above, below, left and right help us describe where a number is. Use your finger to follow a row before you answer. A teacher may also show an unusual number grid: first read its labels, because the pattern may be different.

Worked examples

1. On a normal 1-100 chart, right of 27 is 28, left of 27 is 26. Below 27 is 37; above 27 is 17.
2. To find 10 more than 52, go one row down: 62. To find 10 less than 52, go up: 42.

Hands-on learning

1. Draw a small chart showing 21 to 40 in two rows of ten. Circle 26 and find the numbers on each side.
2. Let a friend call out a number from 1 to 100. Find it quickly on a hundred chart.

Practise in your exercise book

1. What is to the right of 46 on a normal chart?
2. What is immediately before 80?
3. What is ten more than 35?
4. What is ten less than 63?
5. Which is below 18 on a standard chart?
6. Which is above 74 on a standard chart?

Think further

Explain why the number below 41 is 51 on a normal hundred chart.

Try this at home

Make a number chart using a ruler and paper. Ask an adult to check your numbers.

Quick self-check

1. What is 10 more than 29? A. 30 B. 39 C. 19
2. What is just left of 52? A. 51 B. 62 C. 53

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 03 • NUMBER

Measuring by counting handspans and cups

NaCCA indicator: B1.1.1.1.3

I can: Measure length and container capacity by counting repeated informal units.

Let’s learn

We can use body parts and familiar objects to make a first measurement. A handspan is the distance between your thumb and little finger when stretched. A footstep, paper strip or pencil can also help us measure a distance.

To measure a desk with handspans, place one handspan after another, without gaps or overlaps. Count each span. You may find the desk is about six handspans long.

We can compare how much two containers hold using the same small cup. Fill and pour carefully, counting the cups used. The small cup is a measuring unit in this activity.

Measurements made using different-sized hands or cups can be different. This is why we should always say which unit we used and try to use the same unit throughout a comparison.

Worked examples

1. The door is 8 handspans wide using Kojo’s hand: write “8 Kojo handspans”. Do not write simply “8”; tell the unit.
2. A bucket holds 5 small cups of clean water. Another holds 7 of the same cups. The second bucket has greater capacity.

Hands-on learning

1. Measure two tables using a paper strip. Record the number of strips along each table.
2. With an adult, use one cup to compare the capacity of two bowls. Clean spills immediately.

Practise in your exercise book

1. A table takes 6 equal handspans. What is its length in handspans?
2. A container holds 4 cups. Another holds 7 equal cups. Which holds more?
3. Can two children get different handspan measurements of one table?
4. Why must a cup be the same size each time?
5. What should we avoid between repeated paper strips?

Think further

Explain why “my desk is six” is not a complete measurement.

Try this at home

Measure a safe object with pencils laid end to end. Write the count and the unit.

Quick self-check

1. Which is a non-standard length unit? A. handspan B. kilogram C. minute
2. Why use the same cup? A. to be fair B. to use less water C. to finish early

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 04 • NUMBER

More, less and ordering numbers

NaCCA indicator: B1.1.1.1.4

I can: Compare numbers up to 100, find one more or less, and order numbers on a line.

Let’s learn

When one group has more objects than another, its number is greater. When it has fewer objects, its number is smaller. Groups have the same number if every item in one group can be paired with one item in the other group.

Place value helps us compare two-digit numbers. In 47, the 4 stands for four tens; in 52, the 5 stands for five tens. Since five tens are more than four tens, 52 is greater than 47.

On a number line, numbers increase as we move right. When we order from smallest to largest, we use increasing order. From largest to smallest is decreasing order.

A number that is one more than 63 is 64; one less is 62. Use counters or a chart whenever you are unsure.

Worked examples

1. Arrange 15, 9 and 20 in increasing order: 9, 15, 20. In decreasing order: 20, 15, 9.
2. Compare 34 and 43. Both have two digits, but 43 has four tens and 34 has three tens. Therefore, 43 is greater than 34.

Hands-on learning

1. Make two small groups of stones and match stones in pairs to see which has more.
2. Write five different numbers below 100 and ask a partner to arrange them in order.

Practise in your exercise book

1. Which is larger: 56 or 65?
2. Write one more than 79.
3. Write one less than 40.

4. Arrange 7, 19, 13 from smallest to largest.

5. Which is smaller: 28 or 23?

6. Complete: 46, 47, __, 49.

Think further

Explain, using tens, why 81 is greater than 79.

Try this at home

Compare two collections of safe items and tell a family member which has more.

Quick self-check

1. Which is one less than 60? A. 59 B. 61 C. 50

2. Which is smallest? A. 72 B. 27 C. 70

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 05 • NUMBER

Greater than, less than and equal to

NaCCA indicator: B1.1.1.1.5

I can: Use >, < and = correctly to compare numbers.

12 < 19

Twelve is less than nineteen.

Let’s learn

Mathematicians use special symbols so that a number comparison is easy to read. The symbol > means “greater than”. The symbol < means “less than”. The symbol = means “equal to”.

Read the statement from left to right. For example, 8 > 5 is read “eight is greater than five”. The open side of the comparison symbol faces the larger number.

When two numbers are the same, we use the equals sign. Thus 14 = 14. We can check a comparison with number objects, a number chart or a number line.

First decide which number is greater. Then choose a symbol. Take care with 43 and 34: the digits are the same but their places are different.

Worked examples

1. 17 < 21 because 17 is smaller than 21. 90 > 89 because 90 is larger than 89.
2. Six mangoes compared with six mangoes: 6 = 6. The equal sign says the amounts are the same.

Hands-on learning

1. Make large >, < and = symbols using strips of paper. Hold up the correct symbol for two numbers.
2. Ask a partner to check five comparisons using a hundred chart.

Practise in your exercise book

1. Complete: 12 __ 19.
2. Complete: 30 __ 30.
3. Complete: 41 __ 14.
4. Is 6 < 9 true?
5. Is 23 > 32 true?
6. Write a correct comparison for 70 and 60.

Think further

Give two different number pairs that make true “less than” statements.

Try this at home

Find two numbers on a page in a book. Compare them with one of the three symbols.

Quick self-check

1. Choose: 11 __ 15 A. > B. < C. =
2. Choose: 28 __ 28 A. < B. = C. >

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 06 • NUMBER

Comparing numbers in everyday stories

NaCCA indicator: B1.1.1.1.6

I can: Describe differences and order groups using objects, words and a number line.

Let’s learn

Comparing numbers helps us make choices. We might compare the number of chairs in two rooms, the number of pupils in two groups or fruits on two stalls.

We can say a number is a little greater or a lot greater. For example, 25 is only 2 more than 23, while 75 is 52 more than 23. The distance between numbers tells us how different they are.

One-to-one matching is useful when comparing small groups. Put one counter from Group A beside one from Group B until one group runs out. Count the extra counters.

Explain your reasoning in a complete sentence: “Group B has more because three objects are left after matching.” We practise talking about numbers, not only writing answers.

Worked examples

1. Yaw has 8 crayons and Abena has 5. Yaw has 3 more crayons. Difference: count from 5 to 8: 6, 7, 8.
2. Numbers 12, 15, 19: 15 is between 12 and 19. 19 is the largest.

Hands-on learning

1. Build groups of 8 and 11 counters. Match them and say how many extra there are.
2. Draw a number line from 0 to 20; show two numbers and say which is farther right.

Practise in your exercise book

1. How many more is 9 than 6?
2. Which is between 20 and 30: 18, 25 or 35?
3. Put in decreasing order: 6, 14, 10.
4. Which is closer to 40: 39 or 30?
5. What is five more than 13?

Think further

Give two different ways to show that 17 is greater than 12.

Try this at home

Compare the number of spoons and plates at home without moving sharp or hot objects.

Quick self-check

1. How many more is 12 than 10? A. 1 B. 2 C. 3

2. What is between 7 and 9? A. 6 B. 8 C. 10

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 07 • NUMBER

Understanding addition and subtraction

NaCCA indicator: B1.1.2.1.1

I can: Show addition as joining and subtraction as taking away within 20.



5 + 3 = 8

Let’s learn

We add when we put groups together. Addition helps us find how many there are altogether. The plus sign + means add. If you have 5 groundnuts and receive 3 more, you have 8.

We subtract when objects are taken away or when we find how many are left. The minus sign - means subtract. If you have 9 pencils and give away 4, then 5 remain.

A number sentence records our thinking. For example, 5 + 3 = 8 and 9 - 4 = 5. We should always think about the story before choosing addition or subtraction.

Sometimes adding is called joining or increasing. Subtracting is sometimes called separating, taking away or finding a difference. Use pictures or bottle tops to check.

Worked examples

1. Ama has 6 books and gets 2 more: 6 + 2 = 8 books. Because she receives more, we add.

2. Kojo has 12 oranges and gives 5 away: 12 - 5 = 7 oranges. Because he gives some away, we subtract.

Hands-on learning

1. Act out an addition story with 4 and 3 counters. Draw it, then write a number sentence.

2. Use 10 counters to act out a subtraction story and explain what is left.

Practise in your exercise book

1. 4 + 3 = __

2. 10 - 6 = __

3. You had 5 sticks and got 5 more. Add or subtract?

4. You had 8 cups and put 3 away. Add or subtract?

5. 7 + 2 = __

6. 11 - 1 = __

Think further

Make one story that uses 13 - 4, then show it with a drawing.

Try this at home

Tell a family member an addition story with objects in the house.

Quick self-check

1. Which operation means joining? A. addition B. subtraction C. ordering

2. 7 - 2 = ? A. 5 B. 7 C. 9

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 08 • NUMBER

Understanding equal groups and missing numbers

NaCCA indicator: B1.1.2.2.1

I can: Explain equality and represent unknown numbers within addition and subtraction problems.



5 + 3 = 8

Let’s learn

The equals sign = means that what is on the left has the same value as what is on the right. Think of a balanced scale: the amounts on both sides must match.

The statement 3 + 2 = 5 is true because joining three and two makes five. The statement 4 = 2 + 2 is also true. The equal sign does not mean “write the answer next”; it means both sides are equal.

An empty box can stand for a number we do not know yet. For example, 6 + [ ] = 10. We count from 6 to 10: 7, 8, 9, 10. Four steps are needed, so the unknown number is 4.

We can check unknown-number questions using counters. Put out six, add more until you have ten, then count how many you added.

Worked examples

1. 8 = 5 + 3 because each side is eight. 8 = 4 + 4 is also true.

2. 10 - [ ] = 7. Remove 3 from 10 to leave 7. The missing number is 3.

Hands-on learning

1. Make two piles of counters with the same total but different groupings.

2. Draw a box for the missing number in a number sentence and let a partner solve it.

Practise in your exercise book

1. 4 + __ = 9

2. 12 - __ = 10

3. Is 7 = 4 + 3 true?

4. Is 8 = 4 + 5 true?

5. __ + 6 = 11

6. 9 = __ + 1

Think further

Write two different addition expressions that both equal 12.

Try this at home

Use bottle tops to show 7 + [ ] = 13 and check the answer.

Quick self-check

1. Which is equal to 9? A. 4 + 5 B. 3 + 5 C. 9 + 1

2. Find the missing number: 5 + __ = 8 A. 2 B. 3 C. 4

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 09 • NUMBER

Addition and subtraction are connected

NaCCA indicator: B1.1.2.2.2

I can: Use addition to check subtraction and subtraction to check addition within 20.

Let’s learn

Addition and subtraction are related. If 8 + 4 = 12, then 12 - 4 = 8 and 12 - 8 = 4. All three sentences describe the same three numbers.

When we subtract, we can sometimes ask an addition question instead. To find 14 - 9, ask: “What do I add to 9 to make 14?” The answer is 5.

This is useful for checking our work. If we think 13 - 6 = 7, check with 6 + 7 = 13. Since the check is correct, the subtraction makes sense.

We call these related sentences a fact family. Draw three dots or circles with the related numbers and write the addition and subtraction facts using them.

Worked examples

1. 3 + 5 = 8; 8 - 5 = 3; 8 - 3 = 5. The numbers 3, 5 and 8 belong together.

2. To solve 17 - 8, ask 8 + __ = 17. The number is 9.

Hands-on learning

1. Choose three number cards that form an addition fact, then write both related subtractions.

2. Use counters to build 6 + 5 = 11. Remove 5 to show 11 - 5 = 6.

Practise in your exercise book

1. 10 - 8 = __

2. Check 15 - 5 = 10 using addition.

3. For 7 + 4 = 11, complete 11 - 4 = __.

4. Complete 12 - 7 = __.

5. If 9 + 6 = 15, what is 15 - 9?

Think further

Make a fact family using 4, 9 and 13.

Try this at home

Teach a friend how to check subtraction with addition.

Quick self-check

1. What is 13 - 7? A. 5 B. 6 C. 7

2. If 5 + 6 = 11, find 11 - 6. A. 5 B. 6 C. 11

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 10 • NUMBER

Making stories from number sentences

NaCCA indicator: B1.1.2.2.3

I can: Invent and solve everyday word problems for addition and subtraction within 20.

Let’s learn

A number sentence can tell a story. We choose real objects, a person and an action that matches the signs in the sentence. The story needs enough information for someone to work out the answer.

For 6 + 3 = 9, we might say: “Kofi has six mangoes. His aunt gives him three more. How many does he now have?” For 10 - 4 = 6, we might say: “Ama has ten pencils. She gives four away. How many remain?”

The important words are not always the same. Sometimes “altogether” means joining; sometimes “left” means taking away. We should think carefully about what happens to the objects.

Check a story by acting it out with counters. If the objects increase, the story should match addition; if they decrease, it should match subtraction.

Worked examples

1. Number sentence: 7 + 5 = 12. Story: Seven children are playing and five more arrive. There are twelve children altogether.

2. Number sentence: 14 - 6 = 8. Story: Fourteen bananas are in a bowl and six are taken. Eight bananas remain.

Hands-on learning

1. In pairs, write one addition story and swap it with another pair to solve.

2. Draw a comic with three panels: first amount, action, final amount.

Practise in your exercise book

1. Create a short story for 4 + 5 = 9.

2. Create a short story for 9 - 2 = 7.

3. Does “three more children join” suggest add or subtract?

4. Does “five fruits are eaten” suggest add or subtract?

5. Write the answer to 6 + 7.

Think further

Write two different stories that both match 8 + 4 = 12.

Try this at home

Ask an adult to tell you a household counting story. Write its number sentence.

Quick self-check

1. If four friends join five friends, how many? A. 1 B. 9 C. 20

2. Which matches “12 apples, 3 eaten”? A. 12 + 3 B. 12 - 3 C. 3 + 12

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 11 • NUMBER

Friendly number facts and doubles

NaCCA indicator: B1.1.2.3.1

I can: Use number partners to 5 and 10, doubles, and one or two more or less.

Let’s learn

Number facts become easier when we see patterns. Numbers that join to make ten are partners: 1 and 9, 2 and 8, 3 and 7, 4 and 6, and 5 and 5.

A double means adding a number to itself. Double 3 is 3 + 3 = 6. Double 5 is 5 + 5 = 10. Doubles help us think about sums nearby.

If you know 5 + 5 = 10, then 5 + 6 is one more, so the answer is 11. We do not have to guess or memorise without understanding.

We can also think about one more, two more, one less and two less. Ten more than 24 is 34; ten less than 24 is 14. A hundred chart can help us.

Worked examples

1. Find a partner for 7 to make 10: 7 + 3 = 10. Partner for 6 is 4.

2. Double 4 means 4 + 4 = 8. One more gives 4 + 5 = 9.

Hands-on learning

1. Make ten counters in two colours. Record every different way to split them into two groups.

2. Play “double it”: a partner says a number from 1 to 10 and you show its double with counters.

Practise in your exercise book

1. Double 6 is __.

2. What goes with 8 to make 10?

3. What goes with 3 to make 5?

4. One more than 15 is __.

5. Two less than 12 is __.

6. Ten more than 31 is __.

Think further

Explain how double 7 can help you solve 7 + 8.

Try this at home

Practise ten-pair cards with a family member for five minutes.

Quick self-check

1. What completes 6 + __ = 10? A. 3 B. 4 C. 5

2. Double 5 is: A. 10 B. 15 C. 25

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 12 • NUMBER

Smart ways to add and subtract within 20

NaCCA indicator: B1.1.2.4.1

I can: Use counting on, counting back, making ten, near doubles and unknown addends.

Let’s learn

To add a small number, start at the first number and count on. For 8 + 3, say 9, 10, 11. It takes three steps, so the answer is 11.

To subtract, count back. For 15 - 4, say 14, 13, 12, 11. You took four steps, so the answer is 11.

Making ten can make a problem easier. For 8 + 5, take 2 from the 5 to make 8 + 2 = 10. Three are left from the 5, so 10 + 3 = 13.

A near double uses a double we know. If 6 + 6 = 12, then 6 + 7 = 13. To find 12 - 7, ask how much we must add to 7 to make 12. The missing number is 5.

Worked examples

1. 9 + 4: make ten using 1 from the 4, then add 3. Answer: 13. 10 + 3 = 13.

2. 16 - 5: count back five or make the related fact 11 + 5 = 16. Answer: 11.

Hands-on learning

1. Make a number line 0-20. Use little jumps to show 7 + 5 and 13 - 4.

2. Find three different ways to make 15 using addition.

Practise in your exercise book

1. 8 + 6 = __

2. 15 - 6 = __

3. 7 + 7 = __

4. 6 + 7 = __

5. 18 - 9 = __

6. 5 + 9 = __

Think further

Solve 9 + 8 in two ways: make ten and use near doubles.

Try this at home

Use 20 counters to demonstrate one subtraction, then explain it aloud.

Quick self-check

1. Find 8 + 5 A. 12 B. 13 C. 14

2. Find 17 - 8 A. 8 B. 9 C. 10

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 13 • NUMBER

One-step word problems

NaCCA indicator: B1.1.2.4.2

I can: Choose a strategy to solve addition and subtraction stories within 20.

Let’s learn

A word problem uses numbers in a story. Read or listen carefully. Ask: What do I know? What do I need to find? Does the total increase, decrease or need comparison?

Show the problem in a way you understand. You can draw circles, use counters, count on or count backwards. Then write a matching number sentence and an answer with its object name.

Some questions give you the total and ask for a missing part. “There are 14 birds. Six are on the roof. How many are in the tree?” means 14 - 6 or 6 + [ ] = 14.

Always check that your answer makes sense. If a child has 8 mangoes and receives more, the answer should not be less than 8.

Worked examples

1. There are 7 girls and 6 boys. Total? 7 + 6 = 13 children. This is joining.

2. A farmer gathered 18 eggs and sold 9. Left? 18 - 9 = 9 eggs. This is taking away.

Hands-on learning

1. With a partner, act out a shopping story using no more than 20 toy objects.

2. Draw a “story strip”: beginning, change, ending. Include the number sentence.

Practise in your exercise book

1. Ayo has 11 beads and gets 4. How many now?

2. There are 17 cups; 8 are washed. How many are not washed?

3. Eight children are inside and six are outside. Total?

4. Abena has 12 crayons, Kwesi has 9. How many more does Abena have?

5. There were 19 oranges; 5 were eaten. Left?

Think further

Invent a story where the missing number comes at the beginning: “[ ] + 4 = 13”.

Try this at home

Solve a household counting problem using a drawing and a number sentence.

Quick self-check

1. Kofi had 10 pencils and lost 3. Left? A. 7 B. 13 C. 10

2. Nine girls and five boys make: A. 14 B. 15 C. 4

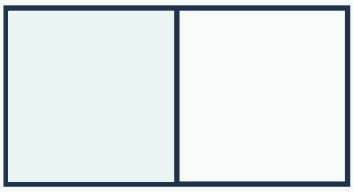
My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 14 • NUMBER

Finding one-half

NaCCA indicator: B1.1.3.1.1

I can: Recognise one-half as one of two equal parts of a whole, without using fraction numerals.



ONE WHOLE

Two equal parts

One part is one-half

Let’s learn

A whole object can be shared. When we divide one whole into two equal parts, each equal part is called one-half.

Equal parts must be the same size. If a loaf is cut into one large piece and one small piece, neither piece is one-half of that loaf. Two halves together make one whole.

We can find halves of paper shapes by folding them so the two parts fit exactly. Some shapes can be folded in more than one way to make equal halves.

Use the words “whole” and “one-half”. In this Basic 1 lesson, we practise recognising and speaking about halves before learning special fraction symbols.

Worked examples

1. Fold a rectangular sheet exactly in the middle. Each equal side is one-half of the sheet. Two equal halves make the whole.

2. A circle split into two equal semicircles shows two halves. A very small slice beside a large slice does not show halves.

Hands-on learning

1. Fold paper circles and rectangles into two equal pieces. Draw a line along the fold.

2. Look at pairs of paper parts and decide whether they are equal.

Practise in your exercise book

1. How many equal parts make one-half?
2. Are a large piece and a tiny piece each one-half of the same whole?
3. How many halves make one whole?
4. Name one object that could be shared into halves.
5. Must two halves be equal in size?

Think further

Draw one rectangle split equally and another split unequally. Explain the difference.

Try this at home

With an adult, share a safe snack equally between two people. Describe each share.

Quick self-check

1. One-half is one of: A. two equal parts B. three equal parts C. two unequal parts
2. Two equal halves make: A. a whole B. three wholes C. nothing

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 15 • NUMBER

Counting halves

NaCCA indicator: B1.1.3.1.2

I can: Count collections of half-pieces using the words one-half, two-halves and three-halves.

Let’s learn

We can count half-pieces just as we count whole objects. One half-piece is one-half. Another equal half-piece makes two-halves, which is one whole.

If we add one more half-piece, we have three-halves. This means one whole and one-half, provided all pieces are halves of equal-sized wholes.

When counting halves, use pieces that are equal in size. A half of a small orange and a half of a large orange are not the same amount, so we must be careful when comparing.

We practise with folded paper, card or drawings. Say the number of half-pieces clearly; we do not need to write fraction symbols yet.

Worked examples

1. One half-piece plus one half-piece = two-halves = one whole. One more half-piece makes three-halves.

2. Four matching half-pieces can make two wholes. Place them in matching pairs to see this.

Hands-on learning

1. Cut or tear paper circles along a fold made by an adult. Pair halves into wholes.

2. Draw three equal half-pieces and say the count: one-half, two-halves, three-halves.

Practise in your exercise book

1. How many half-pieces make one whole?

2. How many half-pieces make two wholes?

3. Say the count for three half-pieces.

4. Two-halves are equal to how many wholes?

5. Why use equal-sized wholes for counting halves?

Think further

Show how five matching half-pieces can form two wholes with one half-piece left.

Try this at home

Fold two equal-sized sheets into halves and count the resulting pieces.

Quick self-check

1. Four matching halves make: A. one whole B. two wholes C. four wholes

2. Three half-pieces are called: A. three-halves B. three wholes C. one-half

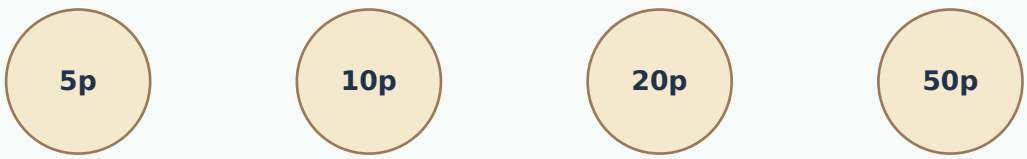
My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 16 • NUMBER

Ghanaian coins and their values

NaCCA indicator: B1.1.4.1.1

I can: Name the Ghanaian coins listed in the curriculum and compare simple amounts in pesewas and cedis.



Paper coin labels for learning values (not real coin images)

Let’s learn

People use money when buying and selling. In Ghana the currency is the Ghana cedi, and one cedi has one hundred pesewas.

The 2019 curriculum introduces one-, five-, ten-, twenty- and fifty-pesewa coins and a one-cedi coin. Some low-value coins may be difficult to find in everyday use. Drawings, cards and photographs can help us learn these values without needing real coins.

A coin has a name and value. Ten pesewas is worth more than five pesewas. Two five-pesewa values make ten pesewas; five twenty-pesewa values make one cedi.

To compare small money amounts, change them to the same unit. For example, one cedi is 100 pesewas, so it is worth more than 50 pesewas.

Always ask an adult before handling money outside class; do not spend money without permission.

Worked examples

1. Two 10-pesewa values together make 20 pesewas. Five 20-pesewa values make 100 pesewas, or 1 cedi.
2. 50 pesewas + 50 pesewas = 100 pesewas. This equals 1 cedi.

Hands-on learning

1. Make labelled paper coin cards with values 1p, 5p, 10p, 20p, 50p and GH¢1. Use them for matching games.
2. Pretend to buy a pencil costing 20 pesewas using two paper cards worth 10 pesewas each.

Practise in your exercise book

1. How many pesewas make one cedi?
2. Which is worth more: 20p or 50p?
3. What is 10p + 10p?
4. How many 50p values make one cedi?
5. What is 5p + 5p?
6. What is 20p + 20p?

Think further

Find two different collections of paper coin cards worth 100 pesewas.

Try this at home

Ask an adult to show you an image of a Ghanaian coin. Read its value without buying anything.

Quick self-check

1. One cedi equals: A. 10 pesewas B. 100 pesewas C. 50 pesewas

2. Which has more value? A. 20p B. 50p C. 10p

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 17 • ALGEBRA

Repeating patterns around us

NaCCA indicator: B1.2.1.1.1

I can: Find, continue, correct and create repeating patterns with two to four elements.



circle, square, circle, square ...

Let’s learn

A pattern repeats in a rule. You may see repeated shapes on a cloth, beads on a string, drum sounds or movements during a game.

A repeating unit is the part that keeps coming again. For red, blue, red, blue, the unit is red, blue. For clap, stamp, turn, clap, stamp, turn, the unit has three actions.

Look carefully before deciding what comes next. Find the unit and say it aloud. You can also spot errors: in a pattern red, blue, red, red, the fourth colour should be blue.

Patterns can have two, three or four items in a repeating unit. Repeating patterns are different from growing number sequences; today we practise the repeat.

Worked examples

1. Circle, square, circle, square, __. The next shape is circle. The repeating unit is circle, square.

2. 1, 2, 3, 1, 2, 3, __. The next number is 1. The unit has three numbers.

Hands-on learning

1. Create a two-colour paper strip inspired by repeating arrangements on Ghanaian woven fabric, without copying a protected design.

2. Make a sound pattern: clap, clap, tap; repeat it three times.

Practise in your exercise book

1. Complete: A, B, A, B, __.

2. Complete: red, red, blue, red, red, __.

3. Complete: 2, 4, 6, 2, 4, __.

4. How many items in the repeating unit: star, circle, triangle?

5. Which repeats next? clap, tap, clap, tap, __.

Think further

Create one pattern with four repeating elements and ask a friend to check it.

Try this at home

Look for repeating shapes on clothes or floor tiles. Draw an original pattern.

Quick self-check

1. Finish: A B C A B C A __ A. A B. B C. C

2. A repeating pattern: A. has a rule B. must grow forever C. has no order

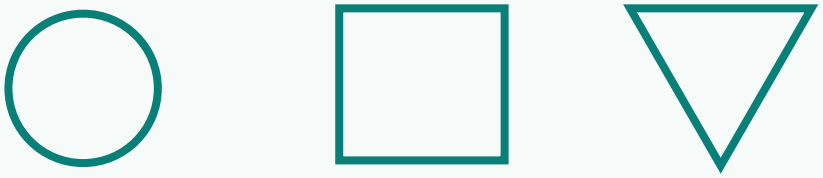
My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 18 • GEOMETRY AND MEASUREMENT

How we know a shape

NaCCA indicator: B1.3.1.1.1

I can: Recognise shape-defining features and distinguish flat shapes from solid objects.



circle square / rectangle triangle

Let’s learn

A flat or two-dimensional shape is like a drawing on paper. A solid or three-dimensional shape is an object that takes up space. A drawing of a ball is flat, but a real ball is a solid object.

A triangle is a closed shape with three straight sides and three corners. A rectangle has four straight sides and four corners. A square is a special rectangle with four equal sides. A circle has one curved boundary and no corners.

Colour and size do not change a shape’s name. A small green triangle is still a triangle. A triangle turned upside down is still a triangle.

We can find flat shapes on faces of solid objects: the flat end of a tin may be a circle, while the face of a box may be a rectangle.

Worked examples

1. Turn a paper triangle around: it remains a triangle because it still has three straight sides. Shape depends on properties, not direction.
2. The flat face of a cube is a square; the whole cube is a solid object. A solid object has depth.

Hands-on learning

1. Draw triangles in three different sizes and point to their three corners.
2. Find one flat face on a safe box or container and trace it on paper.

Practise in your exercise book

1. How many sides does a triangle have?
2. Does turning a rectangle make it a new shape?
3. What makes a square a square?
4. Is a ball a flat or solid object?
5. Does a circle have corners?

Think further

Explain why a blue square and a red square are both squares.

Try this at home

Look for circles, rectangles and triangles in a safe room and sketch them.

Quick self-check

1. A triangle has: A. 3 sides B. 4 sides C. 0 sides
2. A real football is a: A. flat shape B. solid object C. line

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 19 • GEOMETRY AND MEASUREMENT

Solid shapes in the home and school

NaCCA indicator: B1.3.1.1.2

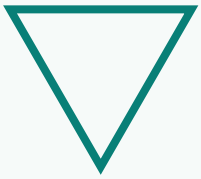
I can: Identify spheres, cones, cylinders, cubes, rectangular prisms and triangular prisms.



circle



square / rectangle



triangle

Let’s learn

Solid shapes have length, width and depth. You can hold most real objects that have a solid shape. Common solid shapes include spheres, cylinders, cones and boxes.

A sphere is round in every direction, like a ball. A cylinder has two matching circular faces and a curved side, like many tins. A cone has one circular base and comes to a point.

A cube is a box with six equal square faces. A rectangular prism is a box shape with rectangular faces; some of its faces may be squares. A triangular prism has two triangular faces and three other side faces.

Compare objects by describing their flat faces, curved surfaces, corners and edges. Be careful: a real-world object may resemble a shape without being perfect.

Worked examples

1. A football is approximately a sphere. A can is approximately a cylinder. Objects can resemble mathematical solids.
2. A classroom box is often a rectangular prism. It has flat faces, edges and corners.

Hands-on learning

1. Make a classroom museum using safe boxes, paper cups and balls. Sort objects by their overall shape.
2. Draw an object beside the name sphere, cube, cylinder or cone.

Practise in your exercise book

1. Which solid looks like a ball?
2. Which solid often looks like a food tin?

3. Which solid has a point and one circular base?

4. What is the name of a box with six equal square faces?

5. Which prism has triangular ends?

Think further

Explain two differences between a cylinder and a cube.

Try this at home

Find three safe household objects with different solid shapes and describe them.

Quick self-check

1. A ball is closest to a: A. sphere B. cube C. cone

2. A cube has how many square faces? A. 4 B. 6 C. 8

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 20 • GEOMETRY AND MEASUREMENT

Flat shapes and their properties

NaCCA indicator: B1.3.1.1.3

I can: Identify circles, triangles, rectangles, squares, rhombuses and hexagons.

Let’s learn

Flat shapes are two-dimensional. A shape can be closed and made of straight lines, or can have a curved boundary.

A triangle has three straight sides. A rectangle has four straight sides and four square corners. A square has four equal straight sides and four square corners. Every square is a rectangle, but not every rectangle is a square.

A rhombus has four equal sides, but its corners do not all have to be square corners. A hexagon has six sides. A circle has no straight sides or corners.

Learn to count sides and corners carefully. Rotate the paper if you need to; a shape does not change when it is turned. We can combine shapes to make pictures or new designs.

Worked examples

1. Count the sides of a hexagon: 1, 2, 3, 4, 5, 6. The shape is a hexagon.

2. A rectangle can be long and thin or nearly square. A rectangle still has four right-angle corners.

Hands-on learning

1. Use paper cut-outs to sort shapes into groups of three, four, six or zero straight sides.

2. Make a house picture from a triangle roof, rectangle wall and circle sun.

Practise in your exercise book

1. How many sides does a hexagon have?

2. What shape has no corners?

3. How many equal sides does a square have?

4. How many corners does a rectangle have?

5. What four-sided shape has all sides equal?

Think further

Explain why every square fits the definition of a rectangle.

Try this at home

Draw six flat shapes, label them and count each shape’s sides.

Quick self-check

1. A hexagon has: A. 5 B. 6 C. 7

2. Every square is: A. a rectangle B. a circle C. a triangle

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 21 • GEOMETRY AND MEASUREMENT

Position and direction words

NaCCA indicator: B1.3.2.1.1

I can: Describe objects using above, below, beside, inside, outside, left, right, near and far.

Let’s learn

Position words tell where one object is compared with another. A cup can be on a table, a bag can be under a chair and a book can be beside a pencil.

Left and right depend on the way a person is facing. Put your hands out and practise. Above means higher than something; below means lower.

A person can be in front of or behind another person in a line. A ball may be inside a box or outside it. When we give directions, we need to mention which object is our starting point.

Clear directions help people find places safely. We can describe positions in a classroom, house, school compound or simple map.

Worked examples

1. The book is on the chair: the book is above the floor and on the chair. The chair is below the book.
2. Ama sits to the left of Kofi. Left tells the position from a viewpoint.

Hands-on learning

1. With a partner, place an eraser above, below and beside a book. Say each position aloud.
2. Draw a simple classroom and describe the place of a door relative to the board.

Practise in your exercise book

1. The sun is __ the roof in a picture where it is higher.
2. A pencil within a box is __ the box.
3. A bag below the desk is __ the desk.
4. Which word is the opposite of left?
5. Which word is the opposite of near?

Think further

Explain why two people facing each other may disagree about what appears on their left.

Try this at home

Draw three toys in different places and write one position sentence for each.

Quick self-check

1. Opposite of above? A. below B. beside C. inside
2. A shoe inside a box is: A. outside B. inside C. above

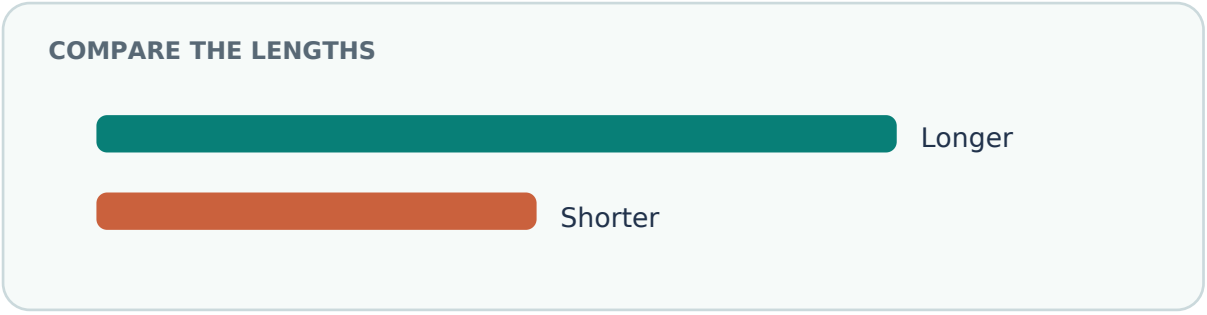
My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 22 • GEOMETRY AND MEASUREMENT

Comparing length, mass and capacity

NaCCA indicator: B1.3.3.1.1

I can: Use everyday language to compare size, length, mass and amount a container can hold.



Let’s learn

Measurement begins with careful comparison. We can say which stick is longer, which child is taller or which ribbon is shorter. Put the ends of two objects together so the comparison is fair.

Mass tells how much matter an object contains; when objects are weighed in the same place, we often say which is heavier or lighter. We can compare safe objects by using our hands or a simple balance made with an adult.

Capacity is how much a container can hold. A large jug may hold more water than a small cup, but we should test with the same measuring cup rather than only judging by height.

Use accurate comparison words: long/short, tall/short, heavy/light, wide/narrow, big/small and holds more/holds less. These words help us explain what we see.

Worked examples

1. A 30 cm pencil is longer than a 10 cm pencil. We compare their lengths.
2. A bucket holding 8 cups has more capacity than a bowl holding 4 of the same cups. We compare their capacities.

Hands-on learning

1. Arrange three ribbons from shortest to longest. Match their ends first.
2. With an adult, compare two safe objects by weight and describe which seems heavier.

Practise in your exercise book

1. Which is longer: a 12-step path or a 5-step path using equal steps?
2. Which is heavier: a full bag of rice or one grain of rice?

3. Which has greater capacity: 7 equal cups or 3 equal cups?
4. What is the opposite of taller?
5. What is the opposite of heavier?

Think further

Find a tall narrow container and a short wide container. Explain why height alone does not tell us capacity.

Try this at home

At home, compare two unbreakable objects by length and report the result.

Quick self-check

1. Which describes capacity? A. how much a cup holds B. how many corners C. how many letters
2. Opposite of heavier: A. taller B. lighter C. longer

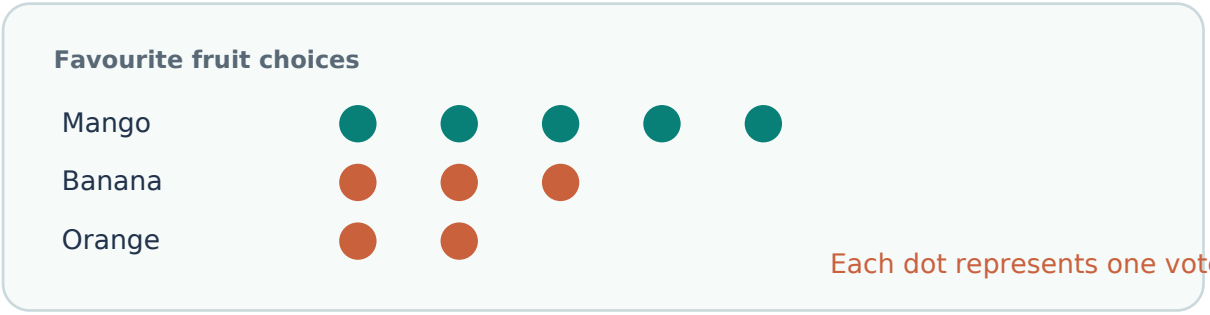
My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 23 • DATA

Making simple picture graphs

NaCCA indicator: B1.4.1.1.1

I can: Organise up to three categories using objects and one-picture-for-one-item graphs.



Let’s learn

Data is information that we collect. We can collect data about favourite fruits, types of transport or pets. To make sense of the data, group similar answers together.

A picture graph shows data using pictures or symbols. In Basic 1, we can let one picture stand for one object. If four children choose mango, we draw four mango symbols in the mango row.

Put category names clearly beside the pictures. Count carefully and compare the rows. The longer row has more pictures when each picture represents the same amount.

A graph should have a helpful title, such as “Our favourite fruits”. Everyone should understand what each picture means.

Worked examples

1. Favourite fruits: mango 4, orange 2, banana 3. Make four, two and three marks in separate rows. Mango has the most.
2. Total choices in this example = 4 + 2 + 3 = 9. Banana has one more choice than orange.

Hands-on learning

1. Survey up to ten classmates about one of three fruits and make a picture graph.
2. Use buttons or counters in rows to represent a prepared set of data.

Practise in your exercise book

1. Mango 5, banana 2, orange 3. Which is most?
2. Which is least?
3. How many choices in total?
4. How many more choose mango than orange?
5. What should a picture graph have?

Think further

Make a graph where two categories have exactly the same number of pictures. Explain what it shows.

Try this at home

Ask five family members to choose one of two safe snacks and make a picture graph.

Quick self-check

1. One picture means one vote. Three pictures mean: A. 1 B. 2 C. 3
2. What helps us know each row? A. category labels B. random numbers C. nothing

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

LESSON 24 • DATA

Counting, tallying and reading data

NaCCA indicator: B1.4.1.1.2

I can: Sort a set of data into three categories, count totals and compare groups.

Let’s learn

A tally mark is a quick line we write each time we count an item. We can draw one short stroke for each object and group every five strokes for easier counting.

When we sort information into categories, we must decide where each item belongs. For example, a class may record the number of pupils choosing football, skipping or dancing.

First count each group. Then add the group counts to find the total number of responses. Compare any two groups to see which is more or less and by how many.

Check your graph and tally chart together: both should tell the same story. If two children worked on the same data, compare their totals and correct any mistakes.

Worked examples

1. Games: football 6, skipping 4, dancing 5. Total = 6 + 4 + 5 = 15. Football has 2 more votes than skipping.
2. Five tallies can be shown as four vertical marks with one diagonal mark across them. For a first activity, counting five separate strokes is also fine.

Hands-on learning

1. Record the colours of ten classroom objects in up to three categories. Tally and graph the results.
2. Swap your tally chart with a partner and independently check each total.

Practise in your exercise book

1. Sport A 7, B 4, C 3. How many in total?
2. Which category has the most?
3. How many more in A than C?
4. How many tallies for 5 objects?
5. If two categories each have 4, which is larger?

Think further

Invent a three-category chart with a total of 12 and explain its different counts.

Try this at home

Keep a tally of three kinds of fruit seen during a safe visit to a market with an adult.

Quick self-check

1. 3 + 4 + 2 data points make: A. 8 B. 9 C. 10

2. If blue has 6 and green has 4, the difference is: A. 2 B. 4 C. 10

My learning reflection: I can explain the lesson to another person. ☐ Not yet ☐ With help ☐ By myself

Reference chart: Numbers 1 to 100

Find a number, then move one space right to add one; move down a row to add ten. Put a finger on a number and practise counting forwards and backwards.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Try it: find 27, 47 and 97; point to the number one more and one less than each.

Answer guide and self-check explanations

The answers are examples or expected responses. Children may show a correct answer in a different way. Encourage explanation, not guessing.

Lesson 01: Counting and naming numbers to 100

1. 40

2. 10

3. 40

4. 18

5. fifth

6. 8

7. 0

Quick check 1: A. 60. After 59 we say 60.

Quick check 2: B. 8. Counting by twos adds 2 each time.

Quick check 3: A. third. Third tells a position.

Lesson 02: Finding numbers on a hundred chart

1. 47

2. 79

3. 45

4. 53

5. 28

6. 64

Quick check 1: B. 39. Moving down one row adds ten.

Quick check 2: A. 51. One step left means one less.

Lesson 03: Measuring by counting handspans and cups

1. 6 handspans

2. The container holding 7 cups

3. Yes

4. For a fair comparison

5. Gaps and overlaps

Quick check 1: A. handspan. A handspan is an informal length unit.

Quick check 2: A. to be fair. Using one cup size makes comparisons fair.

Lesson 04: More, less and ordering numbers

1. 65

2. 80

3. 39

4. 7, 13, 19

5. 23

6. 48

Quick check 1: A. 59. 60 minus one is 59.

Quick check 2: B. 27. 27 has only two tens.

Lesson 05: Greater than, less than and equal to

1. <

2. =

3. >

4. Yes

5. No

6. 70 > 60

Quick check 1: B. <. 11 is less than 15.

Quick check 2: B. =. Both numbers are equal.

Lesson 06: Comparing numbers in everyday stories

1. 3

2. 25

3. 14, 10, 6

4. 39

5. 18

Quick check 1: B. 2. 12 is 2 more than 10.

Quick check 2: B. 8. Eight follows seven and comes before nine.

Lesson 07: Understanding addition and subtraction

1. 7

2. 4

3. Add

4. Subtract

5. 9

6. 10

Quick check 1: A. addition. Addition joins groups.

Quick check 2: A. 5. Count back two from seven.

Lesson 08: Understanding equal groups and missing numbers

1. 5

2. 2

3. Yes

4. No

5. 5

6. 8

Quick check 1: A. 4 + 5. Four plus five equals nine.

Quick check 2: B. 3. Five plus three equals eight.

Lesson 09: Addition and subtraction are connected

1. 2

2. 10 + 5 = 15

3. 7

4. 5

5. 6

Quick check 1: B. 6. Seven plus six equals thirteen.

Quick check 2: A. 5. Undo adding six by subtracting six.

Lesson 10: Making stories from number sentences

1. Example: Four birds came, five joined, nine birds altogether.

2. Example: Nine eggs were in a tray; two were used; seven remained.

3. Add

4. Subtract

5. 13

Quick check 1: B. 9. Five plus four is nine.

Quick check 2: B. 12 - 3. Eaten apples are taken away.

Lesson 11: Friendly number facts and doubles

1. 12

2. 2

3. 2

4. 16

5. 10

6. 41

Quick check 1: B. 4. Six needs four more to make ten.

Quick check 2: A. 10. Five plus five is ten.

Lesson 12: Smart ways to add and subtract within 20

1. 14

2. 9

3. 14

4. 13

5. 9

6. 14

Quick check 1: B. 13. Eight plus two plus three is thirteen.

Quick check 2: B. 9. Eight plus nine equals seventeen.

Lesson 13: One-step word problems

1. 15 beads

2. 9 cups

3. 14 children

4. 3 crayons

5. 14 oranges

Quick check 1: A. 7. Ten minus three is seven.

Quick check 2: A. 14. Nine plus five is fourteen.

Lesson 14: Finding one-half

1. Two equal parts make a whole; one part is one-half

2. No

3. Two

4. Example: orange or paper sheet

5. Yes

Quick check 1: A. two equal parts. A half is one of two equal parts.

Quick check 2: A. a whole. Two halves rejoin to form one whole.

Lesson 15: Counting halves

1. 2

2. 4

3. three-halves

4. 1 whole

5. For a fair comparison

Quick check 1: B. two wholes. Each pair of halves makes one whole.

Quick check 2: A. three-halves. We count the half-pieces.

Lesson 16: Ghanaian coins and their values

1. 100

2. 50p

3. 20p

4. 2

5. 10p

6. 40p

Quick check 1: B. 100 pesewas. There are one hundred pesewas in one cedi.

Quick check 2: B. 50p. Fifty pesewas is the greatest value shown.

Lesson 17: Repeating patterns around us

1. A

2. blue

3. 6

4. 3

5. clap

Quick check 1: B. B. The pattern is A B C repeated.

Quick check 2: A. has a rule. A clear repeating unit follows a rule.

Lesson 18: How we know a shape

1. 3

2. No

3. Four equal straight sides and four right-angle corners

4. Solid

5. No

Quick check 1: A. 3 sides. A triangle has three straight sides.

Quick check 2: B. solid object. It occupies space in three dimensions.

Lesson 19: Solid shapes in the home and school

1. Sphere

2. Cylinder

3. Cone

4. Cube

5. Triangular prism

Quick check 1: A. sphere. A sphere is round like a ball.

Quick check 2: B. 6. It has six square faces.

Lesson 20: Flat shapes and their properties

1. 6

2. Circle

3. 4

4. 4

5. Rhombus (a square is a special rhombus)

Quick check 1: B. 6. Hexa means six in this shape name.

Quick check 2: A. a rectangle. A square has four right-angle corners.

Lesson 21: Position and direction words

1. above

2. inside

3. under / below

4. right

5. far

Quick check 1: A. below. Below means lower than.

Quick check 2: B. inside. Inside means within the box.

Lesson 22: Comparing length, mass and capacity

1. 12-step path

2. The full bag

3. 7-cup container

4. shorter

5. lighter

Quick check 1: A. how much a cup holds. Capacity is how much a container holds.

Quick check 2: B. lighter. Lighter means less heavy.

Lesson 23: Making simple picture graphs

1. Mango

2. Banana

3. 10

4. 2

5. A title and labelled categories

Quick check 1: C. 3. Count one per picture.

Quick check 2: A. category labels. Labels tell what the rows mean.

Lesson 24: Counting, tallying and reading data

1. 14

2. A

3. 4

4. 5

5. They are equal

Quick check 1: B. 9. Add the category totals.

Quick check 2: A. 2. Six minus four equals two.

End of guide

You have practised the Basic 1 mathematics strands. Keep using numbers when shopping with an adult, observing shapes, playing pattern games, measuring household objects and collecting simple data. Mathematics grows stronger with regular practice.